Neuro-Evolved Heuristics for Variable Gapped Common Subsequence Identification

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Introduction

- Objects we deal with: sequences (strings) over finite alphabet Σ
  - DNA/RNA sequences over $\{\mathtt{A}, \mathtt{T}, \mathtt{G}, \mathtt{C/U}\}$
  - Proteins over 20 (canonical) amino acids: $\{\mathtt{A}, \mathtt{C}, \mathtt{D}, \mathtt{E}, \mathtt{P}, \mathtt{Q}...\}$
- **One of central tasks in computational biology:**
  - sequence comparison, finding **common motifs** between sequences
  - how structural similarity affects sequence functionality
- Motifs can be **subsequences**: **Longest commmon subsequence problem** variants (studied 50 years already)

Longest common subsequence problem (LCSP)

Definition (LCSP)

Input: Given an arbitrary set of sequences $S = \{s_1, \ldots, s_m\}$
Task: Find a subsequence s common for all sequences from S with maximum possible length ($|s|$).

LCS: Literature & Problem Variants

- For $m = 2$, polynomially solvable, e.g. Dynamic programming
- When m arbitrary large, it is $\mathcal{NP}$-hard: approximation, meta-heuristic, exact approaches

Problem-related practical variants: Repetition-free, Filled LCS problem, Gapped LCS problem

The gapped LCS problem (Peng and Yang, 2012)

Definition (Gapped sequence)

Given is a sequence s and a function $G_s : \{1, \ldots, |s|\} \mapsto \mathbb{N}$. A pair (s, G_s) is called a gapped sequence.

(G_s: To each position of seq. s, one number assigned)

Definition (Gapped subsequence)

Sequence $\tilde{s}$ is a gapped subsequence of (s, G_s) iff

- $\tilde{s}$ is a **subsequence** of s
- the gap constraint G_s fulfilled resp. the *positions of embedding* $\tilde{s}$ in s:
  - consecutive embedding positions respect gap distances (no more than the assigned number of letters inbetween respected)

Problem definition

Example

$s = \texttt{AATTGC}$, $G_s(\cdot) = 1$, $\tilde{s} = \texttt{ATG}$

- the embedding AATTGC (valid gapped subsequence)
- the embedding AATTGC (invalid gapped subsequence)

Definition (The multiple (variable) gapped LCS problem – MVGLCSP)

Input: Given is a set of m gapped sequences $\{(s_i, G_{s_i})\}_{i=1}^{m}\}$.
Task: Find the longest sequence $\tilde{s}$ so that

- $\tilde{s}$ is **common subusequence** of each s_i
- $\tilde{s}$ **fulfills all gap constraints** G_{s_i} $(i = 1, \ldots, m)$

Literature & Motivation for VGLCSP

- Peng and Yang (2012, 2014): $m = 2$ version by three dynamic programming (DP) approaches
- Manea et al. (2024): complexity analysis investigated
  - $\mathcal{NP}$-hard under arbitrary large m
- Djukanovic et al. (2026) proposed the first approach for m arbitrary: Iterative Beam search approach

Motivation:

- Genetics and molecular biology: DNA/protein comparison analysis–distances between residues must be respected
- Time-series analysis: in settings where events are required to occur within specified temporal delays (Lainscsek et al. (2015))

Methodology by Djukanovic et al. (2026)

- Based on the **state space graph formulation** of the problem
- **Nodes** corresp. to partial solutions and resp. subproblems to be solved
  - transitions: valid extensions of partial solutions (by one letter)
- **Gap constraints**: incorporated to cut-off invalid extensions (edges) among the LCS extensions immediately
- **Many root/source nodes** in the state space—the position of leading letter in each common gapped subsequence free to select
  - root state space subgraph (formulation) w.r.t. starting source (match)

Methodology: main components roughly

- Generation of longest solution for a root (sub-)space is generally hard $\Rightarrow$ need an approx. approach for evaluation (**Beam search** – **limited BFS**)

- The root-state-space formulation may involve disconnected components generation $\Rightarrow$ needs **iterative beam search** to explore all space

$\Longrightarrow$ **Iterative Multi Source Beam Search (IMSBS) approach**.

Heuristic guidances in IMSBS

The best literature heuristic to guide BS and prioritize root node:

1. UB_2: *Character Frequency Alignment* score

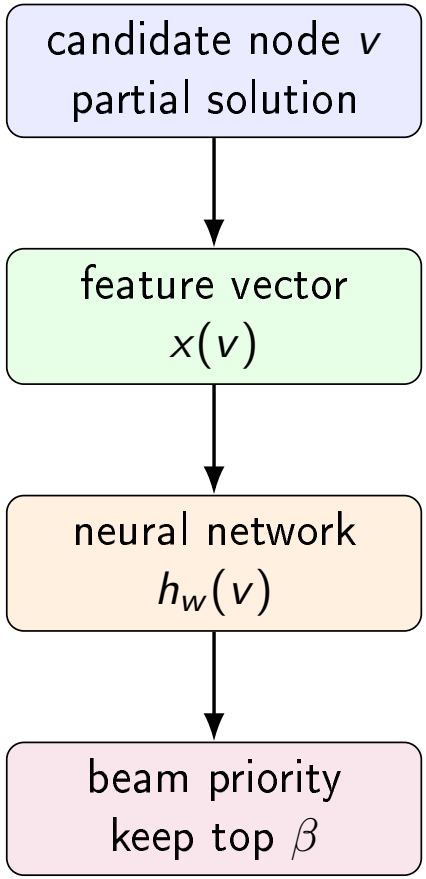
Experimental evaluation:

- IMSBS able to efficiently run on large synthetic instances
- Weaknesses: the performance of UB_2 seems to significantly drop for large n $(= 500)$ and small $|\Sigma|$ $(= 2)$

$\Rightarrow$ **Construction of a new heuristic guidance**

Designing advanced heuristic guidances: main idea

- IMSBS needs a better score to decide which candid nodes stay in the beam
- We learn a new node evaluator: a function assigning a promise score to a partial solution

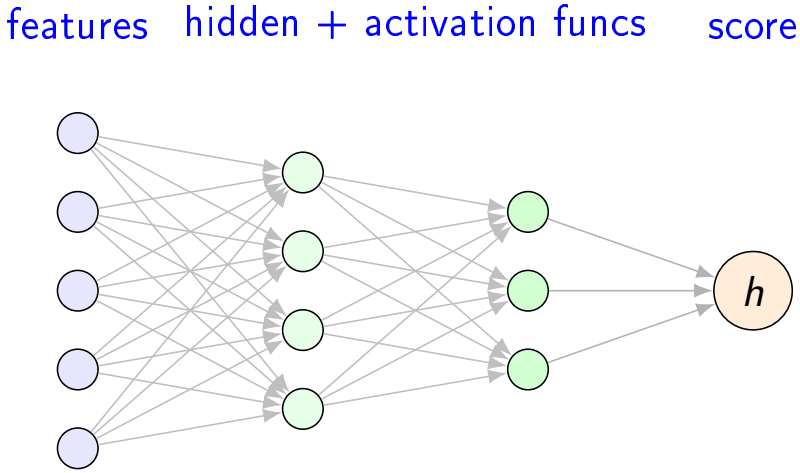


Role of the learned heuristic

It does **not** construct the solution directly; it ranks candidate states during search.

Neural network basics: MLP as a scoring function

- We use a **feed-forward multilayer perceptron** (MLP).
- Input: **compact numerical description** of the state v.
- Hidden layers transform features into nonlinear combinations.
- Output: a **single heuristic score** $h_w(v)$.



State features for VGLCS node evaluation

For a state $\mathbf{v} = (\mathrm{p}^v, l^v)$ (p^v: vector of suffix positions of subproblem to be solved, l^v: length of partial solution), build a size-independent features:

Search progress

- normalize positions: $F_i^v = p_i^v / |s_i|$
- max, min, mean, std of F^v
- current length l^v

Gap flexibility

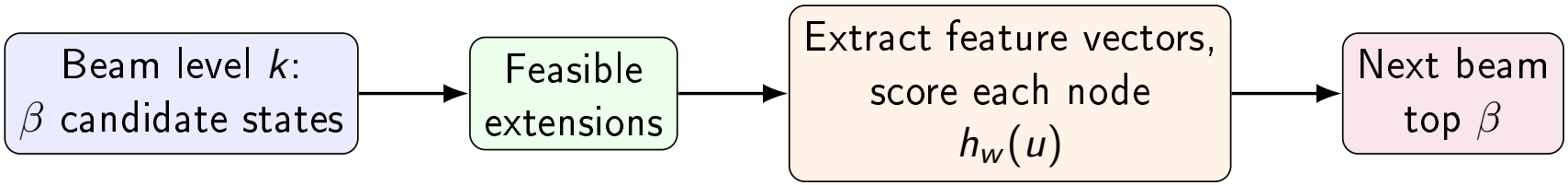
- $G_i^v = G_{s_i}[p_i^v, \ldots]$
- max, min, mean, std
- describes how restrictive future extensions are

Global context

- alphabet size $|\Sigma|$
- number of sequences m
- standardized before training

Feature sets tested: F_1 = progress + gaps (9); $F_2 = F_1 + |\Sigma|$ (10); $F_3 = F_2 + m$ (11).

How the learned evaluator is used inside IMSBS



Important note

Learning is **offline**; during the final run the NN is only a fast scoring function.

Why not standard supervised backpropagation?

Obstacle

- Exact optimal trajectories unavailable for the general case
- Search quality non-differentiable: one ranking decision affect many later states

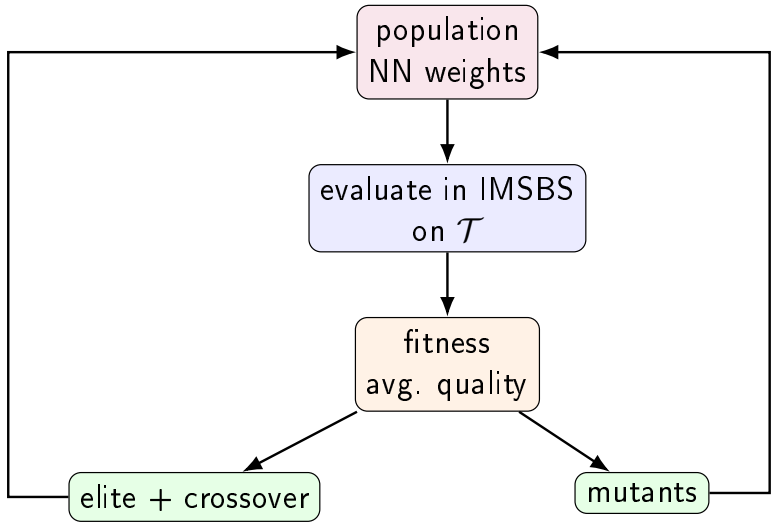
Solution: neuro-evolution

- Treat all NN weights as **one vector** w.
- Evaluate w by running (inexpensive) IMSBS on training (32) instances.
- **Fitness** = average subsequence length found

$$\max_w \frac{1}{|\mathcal{T}|} \sum_{I \in \mathcal{T}} f(\text{IMSBS}(I, h_w))$$

Learning weights with Biased Random Key GA

- Individual = complete NN weight vector w.
- Population of weights, evolution ...



Parameters used

$n_{pop} = 20$, $n_{elite} = 1$, $n_{mutants} = 7$, $p_{bias} = 0.5$.

Rank-based ensemble: learned + analytical guidance

- Preliminary tests: the learned heuristic strong, but not uniformly best
- Hand-crafted upper bounds useful, especially in some small-m cases
- Raw scores not directly comparable, **rankings** rather than scores combined

Combined rank

$$r(v) = \alpha\, r_{h_{w^*}}(v) + (1-\alpha)\, r_{UB_2}(v).$$

Lower rank $r(v)$ means higher priority in the beam.

$\Rightarrow \mathrm{UB}_2$ replaced by $r(\cdot)$: **Ensemble-IMSBS approach**.

What was tuned? Experimental design

IMSBS parameters

- tuned by `irace` (parameters from the literature: $\beta = 5000$

Training protocol

- IMSBS utilized $\beta = 500$, max. 100 beam iterations or 60 sec

NN design choices

- 2 or 3 hidden layers
- 5 or 10 neurons per hidden layer
- activation: `tanh`, `ReLU`, `sigmoid`
- features: F_1, F_2, F_3

Selected architectures (split by diff. m)

m	layers	act.	features
2	10-5-5	ReLU	F_2
3	10-10-5	ReLU	F_2
5	10-5-5	sigmoid	F_1
10	10-5-5	ReLU	F_2

Numerical results: Random and Real benchmarks

Random benchmark

comparison [Ensembe vs. IMSBS]	wins	ties	losses
all groups (32)	**20**	8	4
$m = 2, 3$	**10**	5	1
$m = 5, 10$	**10**	3	3

Wilcoxon: $p = 0.009$ on all groups.

Real benchmark

comparison	wins	ties	losses
LIMSBS-Ensemble vs. IMSBS (20)	**12**	7	1

Comparable runtime; larger gains on harder instances such as Virus.

Takeaway

The learned ensemble improves solution quality consistently, while keeping the original IMSBS machinery almost unchanged.

Conclusion

- We successfully learned an improved guidance (best-performing heuristic) through a neuro-evolved process
  - statistically significantly better results w.r.t. the literature-best approach

- Noticed for $|\Sigma| = 2$: a stronger exploration/**richer feature portfolio** needed

Thank you for your attention!

Ensemble-IMSBS vs. exact DP-1 solution ($m = 2$)

$|\Sigma| = 2$

n	Avg. Gap
50	1.0%
100	6.5%
200	10.0%
500	**24.4%**

Performance degrades as the instance size increases.

$|\Sigma| = 4$

n	Avg. Gap
50	0.7%
100	0.3%
200	1.1%
500	**2.0%**

Near-optimal performance across all instance sizes.

Ensemble-IMSBS stays within 2% of DP-1 for all tested $|\Sigma| = 4$ instances.